

Guilherme Iglesias Rocha Gomes

Mechanical Engineering | *Data & Software Engineering*
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PROFILE

Master's student in Mechanical Engineering at UPB and research fellow at CeDRI in the Biwit project, working with Python, large language models, computer vision and data pipelines. I have been involved in scientific research since 2020, starting with a scientific initiation scholarship in Brazil (two papers at SVR) and continuing with VineEye, my final-year project in agricultural robotics with AI, which won the Douro Interior Prize at the Mostra Nacional de Jovens Empreendedores 2026 in Porto. I have experience building data pipelines with LLMs, structured datasets and property graph data models. Career goal: Software Engineer or Data Engineer, building data and AI systems for industry and agriculture.

TECHNICAL SKILLS

Programming & Data
- Python
- C#
- SQL
- HTML / CSS / Java
- JSON

Engineering & Design
- SolidWorks / AutoCAD
- ANSYS & Fluent Meshing
- Matlab
- 3D Printing and CNC
- Arduino

AI & Data Management
- YOLOv12n
- LLMs
- Mathematical modelling
- Dataset creation

Tools & Standards
- Microsoft Excel
- Unity 3D Engine
- Git / GitHub
- LaTeX / Overleaf

EDUCATION

Master's Degree in Mechanical Engineering | UPB, Polytechnic University of Bragança *Sep. 2026 - Present*
- 1st-year student (EQF Level 7), Bragança, Portugal

Bachelor's Degree in Mechanical Engineering | UPB, Polytechnic University of Bragança *Sep. 2023 - Sep. 2026*
- Completed (EQF Level 6), Bragança, Portugal
- **Final-year project:** VineEye, stereo vision system for vineyard robotics using dual ESP32-CAM

Mechanical Engineering Technician, Integrated Secondary Education | IFES, Vitória, Brazil *Feb. 2019 - Dec. 2022*
- Integrated technical and scientific training (EQF Level 4)
- Foundation in mechanics, automation, programming and industrial projects

WORK EXPERIENCE

Research Fellow (BII) | CeDRI, Polytechnic University of Bragança *Apr. 2026 - Present*
- Biwit project, precision viticulture research line
- Pipeline for collecting Portuguese agricultural news and classifying it with LLMs
- Climate-event detector and yearly reports linking news, climate and market signals to wine production models

Mechanical Engineering Intern | Cut Out, Bragança, Portugal *Jun. 2025 - Dec. 2025*
- Production support in 3D printing, laser cutting and CNC, from preparation to quality control
- 2D and 3D modelling with SolidWorks and AutoCAD
- Process optimisation and team workspace organisation

Private Tutor | Universo do Saber, Bragança, Portugal *Dec. 2023 - Mar. 2025*
- Subjects: Mathematics, Calculus, Algebra and English, 150 teaching hours
- Students ranging from primary school to university level

RESEARCH PROJECTS

Final-Year Project: VineEye, Stereo Vision for Agricultural Robotics | UPB *Nov. 2025 - Sep. 2026*
- Low-cost system with dual ESP32-CAM, Python and YOLOv12n for autonomous navigation in vineyards
- Camera calibration, hardware synchronisation, real-time depth mapping with low-cost material
- Developed with CeDRI and AVIPE
- Presented at IN2COP 2026 International Conference, Bragança
- Exhibited at the Feira Nacional de Agricultura, Santarém, and at the Mostra Nacional de Jovens Empreendedores, Porto, where it won the Douro Interior Prize (Consulting and Mentoring)

Erasmus: University Registration System | Vilniaus Kolegija, Lithuania *Nov. 2025*

- Registration system using SQL, HTML/CSS and Java; relational database design and management

Erasmus: Robotic Arm | Vilniaus Kolegija, Lithuania *May 2025*

- Mechanical design in SolidWorks, 3D prototyping, programming with Arduino and Python

Erasmus: Digital Challenge Innovation Lab (D-Chill) | Hanze University, Groningen, Netherlands *Dec. 2024*

- Educational game to teach Python to children, developed in partnership with SharkCoders
- Participation in IN2COP 2025 International Conference, UPB

Scientific Initiation Scholarship: Industrial Virtual Reality | Vale / IFES, Vitória, Brazil *Nov. 2020 - Mar. 2023*

- VR training simulator for industrial machine inspection (IFES/Vale partnership)
- Serious game in Unity3D with C# and Python; creation of AI training datasets
- Scientific articles presented at SVR 2021 and SVR 2022 symposia
- Instructor in serious game development courses

AWARDS AND COMPETITIONS

Douro Interior Prize (Consulting and Mentoring) | Mostra Nacional de Jovens Empreendedores, Porto *2026*

- Awarded to VineEye, low-cost stereo vision for vineyards

Top 5 of 21 teams | EY AI Challenge, Lisbon *May 2026*

- LusoLaw: legal AI assistant on Portuguese labour and civil law, built in 4 hours by a team of 5 people

PUBLICATIONS

- Oliveira, T. R., **Gomes, G. I. R.**, Martins, T. S., Borges, L. F. M. R., Ludke, G. G., Andreão, R. V., Cotini, L. G., Viana, P. H. P., Schimidt, M. Q., Mestria, M. (2022). Virtual Reality System for Inspection and Training in Wheel Loader. *Proceedings of the Symposium on Virtual and Augmented Reality (SVR 2022)*, 11–20.
- Oliveira, T. R., Silva, M. M., Spinassé, R. A. N., Ludke, G. G., Gaudio, M. R. S., **Gomes, G. I. R.**, Cotini, L. G., Vargens, D., Schimidt, M. Q., Andreão, R. V., Mestria, M. (2021). Systematic Review of Virtual Reality Solutions Employing Artificial Intelligence Methods. *Proceedings of the Symposium on Virtual and Augmented Reality (SVR 2021)*, 42–55.
- **Gomes, G. I. R.** et al. (2026). Extended abstract on VineEye, low-cost stereo vision for vineyards. *IN2COP 2026 International Conference on Co-Creation Processes in Higher Education*, Bragança.

LANGUAGES

Portuguese: Native language **English:** C2 Proficient